ELEOS Adventure

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Background

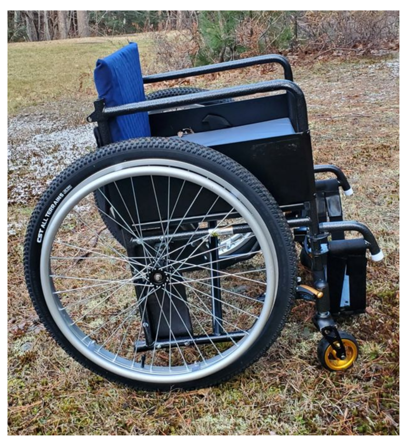
Traditional wheelchairs are impractical for rough terrain. Current solutions are expensive, difficult to manufacture, and have limited mobility. This project aims to create a new design that maximizes off road capability, all while keeping costs low, and using accessible materials.

Existing Solutions

GRIT Freedom Chair

Pros:
- Lever driven
- Large front caster
- Easy assembly

Cons:
- Expensive
- Complex custom-machined parts



Frankie

Pros:
- Modifies existing wheelchairs
- Inexpensive

Cons:
- Marginal offroad improvement w/o trike mod

Final Design

Figure 1. The ELEOS Adventure

Client Specifications

1. A functioning prototype
2. Made from commercially available materials
3. Material costs below $800
4. Design is easy to reproduce

Test Results

Table 1. Time trial results from three different terrains (Data was collected from three distinct participant groups: children, adult males, and adult females.)

Terrain	Adventure (s)	Standard (s)	Improvement
Uphill	54.9	106.4	49%
Dirt	16.6	33.1	50%
Pothole	21.7	24.9	12%

Future Development

Bike Headset
- Lower cost
- Easy to source
- Simplify assembly

Better Disc Brakes
- Improved control
- Shorter stopping distance
- Greater safety

Reinforced Seat Hinge
- Increased lifespan
- Greater durability
- Stronger under load

Motors*
- Higher speed
- Improved performance
- Increased efficiency

*** This would expand the project scope**

Important Features

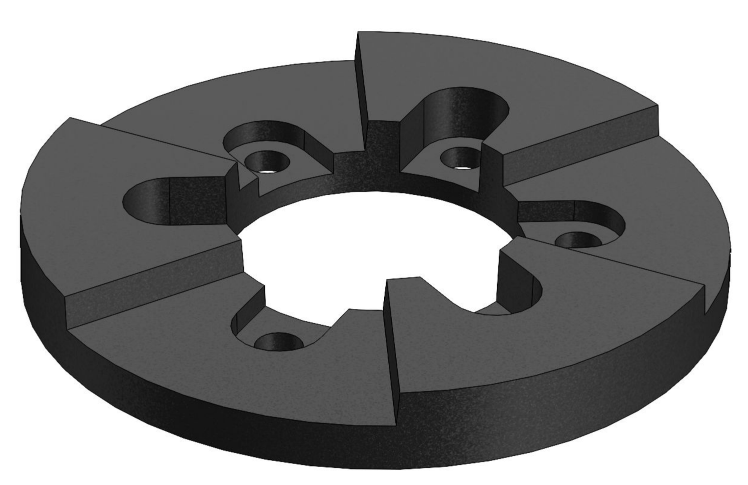


Figure 3. Dog Gear 3D Model

Dog Gears and Quick Release
- Quick wheel removal, making it easy to change wheels or reduce size for storage and transport.
- Nylon selected for its high strength and suitability for injection molding in mass production.

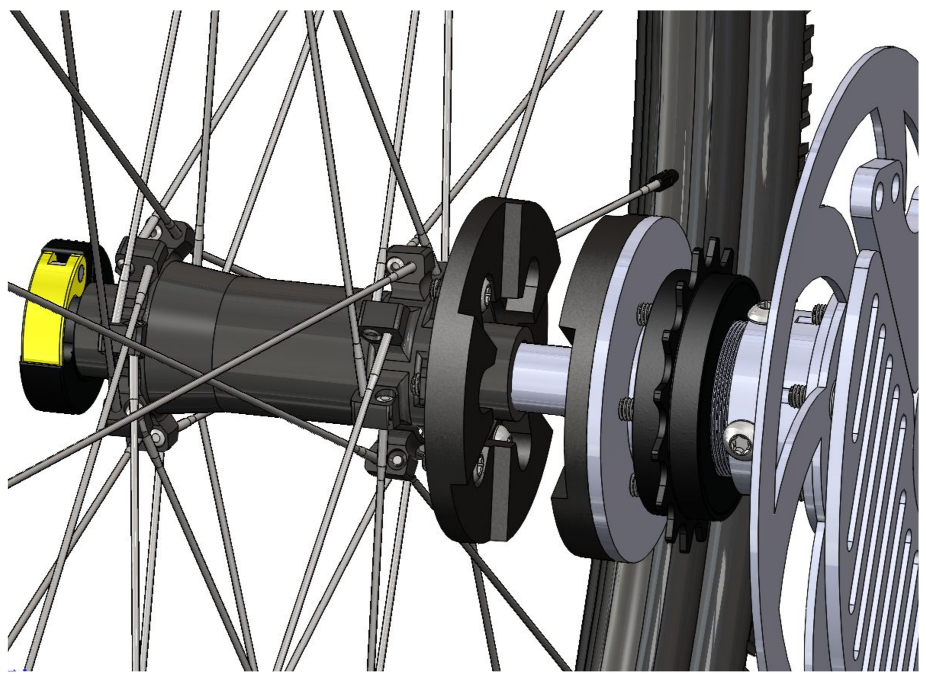


Figure 4. Dog Gear Exploded View

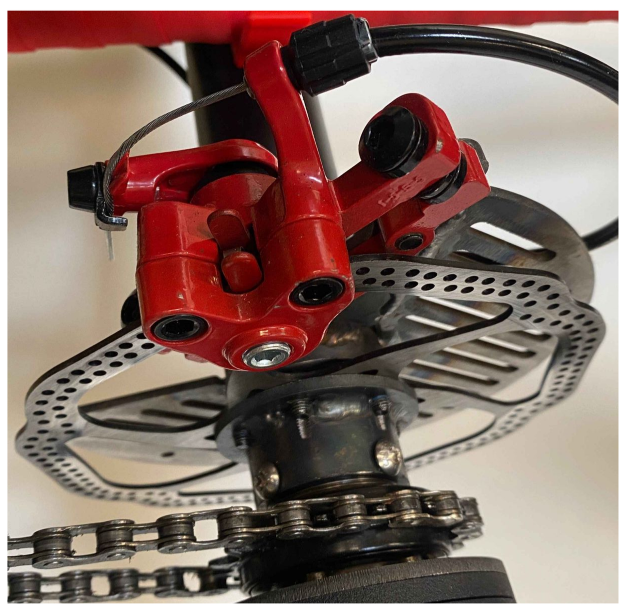


Figure 5. Disc Brakes

Disc Brakes
- Improves hill climbing by preventing roll back between pushes
- Improves user control
- Increases overall safety on diverse terrain
- Eliminates the need for pushrim braking

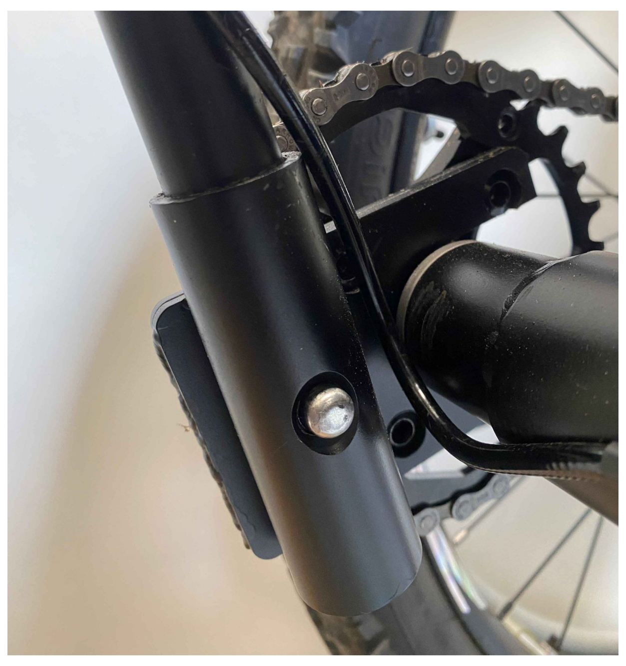


Figure 6. Lever Attachment Mechanism

Detachable Levers
- The levers can be removed using a quick-detach mechanism
- Allows the chair to operate in reverse
- Can be stored using the clips on the frame for quick access or fully removed
- The scrub brakes function as a secondary brake and prevents rolling backward

Figure 7. Wheel with Pushrims and Bike Tires

Bike Tires with Pushrims
- Improves ability to traverse diverse terrain
- Provides better traction on uneven surfaces
- Bike tires increase propulsion efficiency
- Versatile indoor and outdoor use
- Pushrims allow standard manual wheelchair operation

Team Members

Left to Right: Owen Tonseth, Wyatt Bertis, Gus Seigneur, Caylem Spengler, Isaiah Jones